

WEST ORCA BAY BUOY 46060, AK, USA

WMO: 992850

Lat: 60.587N

Lon: 146.819W

Elev: 0

StdP: 101.33

Time Zone: -9.00 (NAK)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.0	-4.8	N/A	N/A	N/A	N/A	N/A	N/A	19.0	3.4	17.4	3.5	5.9	20	0.762

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.0	16.9	N/A	16.1	N/A	15.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.5	13.0	DB	-7.5	18.1	2.7	1.6	-9.5	19.3	-11.1	20.2	-12.6	21.1	-14.6	22.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	1.7	1.4	1.6	3.9	7.8	11.3	13.7	13.4	10.9	7.3	3.6	2.6	
	DBStd	5.04	3.03	2.59	2.36	1.96	2.09	1.89	1.40	1.30	1.57	2.02	2.86	2.51	
	HDD10.0	1533	257	241	261	182	72	8	0	0	9	84	191	229	
	HDD18.3	4268	515	474	519	432	326	212	144	152	224	341	441	488	
	CDD10.0	307	0	0	0	0	4	46	114	107	35	1	0	0	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	5.4	6.2	6.2	5.6	4.9	4.0	3.9	4.0	4.5	6.0	6.5	6.7	7.0	
Precipitation	PrecAvg	1424	138	120	97	74	73	70	97	133	177	155	137	154	
	PrecMax	1832	324	256	243	141	149	115	159	241	303	313	317	304	
	PrecMin	985	30	2	28	24	21	21	52	28	71	77	2	51	
	PrecStd	178	62	54	54	29	35	22	25	56	60	60	74	61	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	6.7	6.2	10.6	14.2	18.0	18.8	18.0	14.9	11.9	9.4	8.0	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	6.4	5.7	5.5	8.7	12.5	16.0	17.2	16.8	14.1	10.9	8.7	7.2	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	5.8	5.1	5.0	7.3	11.4	14.8	16.2	16.0	13.5	10.4	7.7	6.5	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	5.2	4.5	4.5	6.3	10.6	13.8	15.5	15.4	13.0	9.9	7.0	5.7	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.5	2.3	2.6	2.5	2.4	2.2	2.0	2.0	2.0	2.1	2.5	2.5	
		MCDBR	2.6	2.4	2.8	3.4	3.6	3.6	2.9	3.0	2.3	2.3	2.7	2.6	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.269	0.311	0.323	0.363	0.382	0.370	0.384	0.377	0.344	0.323	0.274	0.403	
	taud		2.154	2.276	2.331	2.266	2.288	2.412	2.410	2.421	2.487	2.362	2.149	2.624	
	Ebn at Noon		476	676	803	829	840	860	836	811	781	655	451	307	
	Edh at Noon		40	66	86	110	116	105	103	94	75	60	39	29	
All-Sky Solar Radiation	RadAvg		0.24	0.81	2.17	3.74	4.91	5.19	4.46	3.41	2.27	1.02	0.35	0.14	
	RadStd		0.04	0.19	0.35	0.44	0.55	0.42	0.51	0.53	0.43	0.15	0.09	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page